

AI-BASED INVESTMENT PLAN RECOMMENDATION MODULE FOR DIGITAL WALFARE MANAGEMENT IN SRI LANKA

25-26J-320

Project Proposal Report

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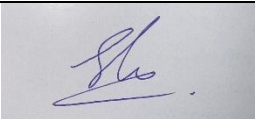
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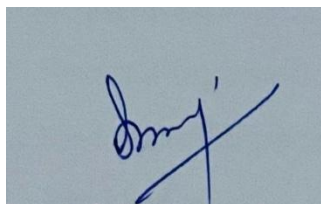
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Declaration

I hereby declare that this proposal is my original work and does not incorporate any material previously submitted for any academic award. All sources have been properly acknowledged.

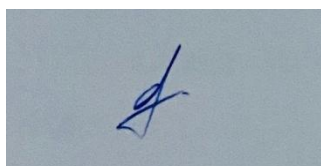
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Abstract

This research focuses on the design and implementation of an AI-Based Investment Plan Recommendation Module aimed at empowering low-income welfare recipients in Sri Lanka. By leveraging machine learning algorithms, the system will analyze user income, expense patterns, and financial risk profiles to recommend safe, practical, and sustainable investment or savings opportunities. The module is part of a larger AI-powered Digital Welfare Platform, ensuring that welfare beneficiaries receive not only immediate financial aid but also guidance for long-term economic stability.

The proposed solution combines clustering techniques to segment users by financial capacity and collaborative filtering to match them with investment schemes aligned to their needs. Recommendations will be personalized, transparent, and compliant with ethical AI principles. Ultimately, this module aims to reduce dependency on welfare, promote financial literacy, and improve recipients' ability to participate in the economy.

Keywords: Financial AI, Recommender Systems, Clustering, Low-Income Empowerment, Digital Welfare Systems.

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1. INTRODUCTION

This research project aims to design and develop an AI-powered Investment Plan Recommendation Module as part of a broader digital welfare management platform for Sri Lanka. The country's social protection system offers multiple programs, including Samurdhi, disability allowances, elderly pensions, and nutritional aid. While these schemes address immediate needs, they often lack mechanisms to guide beneficiaries toward long-term financial independence. Many recipients remain dependent on welfare due to limited financial literacy, lack of tailored guidance, and absence of accessible tools to evaluate safe investment opportunities.

Current welfare administration faces challenges such as fragmented data handling, absence of personalized financial advisory services, and lack of predictive systems for economic empowerment. Internationally, countries have integrated AI-driven financial planning tools into welfare programs to promote self-sufficiency. For example, micro-investment advisory systems in India and Bangladesh have demonstrated significant improvements in household savings and income diversification. However, Sri Lanka has yet to adopt a unified, AI-enabled financial recommendation solution within its welfare ecosystem.

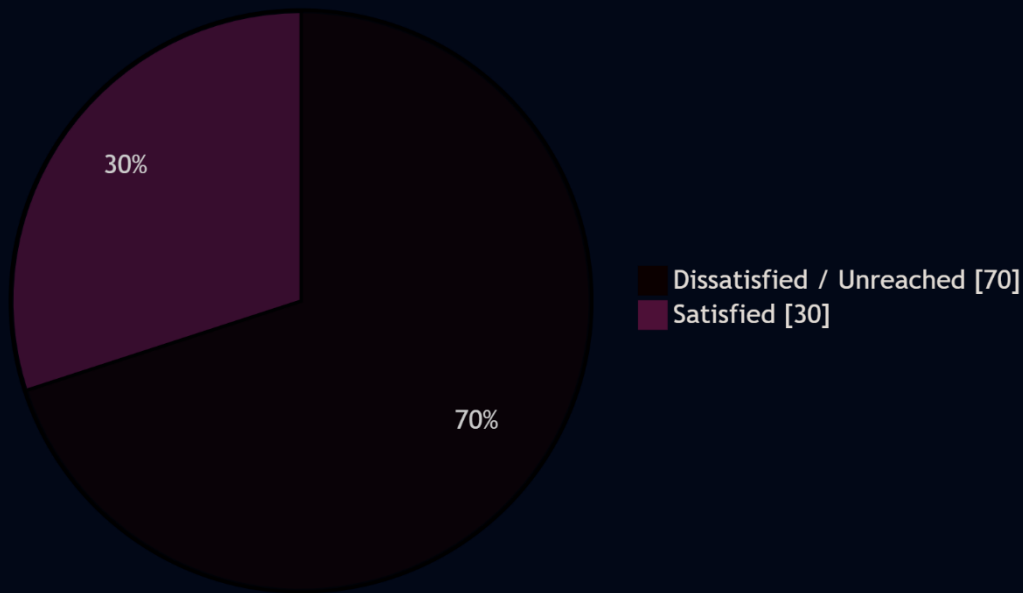
The proposed module is designed to enhance—not replace—existing welfare processes by offering personalized, risk-aware investment guidance to low-income beneficiaries. It uses machine learning algorithms to analyze user profiles, including income, expenses, savings patterns, and risk tolerance, to recommend suitable micro-investment and savings plans. The system will incorporate clustering techniques to segment users by financial capacity and collaborative filtering methods to match them with relevant opportunities from government, NGOs, and regulated financial institutions.

Functioning as part of the integrated welfare platform, this module will benefit three key user groups:

- **Citizens:** Gain accessible, personalized financial recommendations in their preferred language.
- **Officers:** Monitor adoption and outcomes of financial plans for welfare recipients.
- **Policymakers:** Access analytics to evaluate program impact and optimize resource allocation.

By embedding AI-driven investment guidance into the welfare system, this research aims to reduce long-term dependency on aid, promote financial stability, and contribute to sustainable socio-economic development in Sri Lanka.

"Beneficiary Satisfaction (Traditional Methods)"



This component aims to harness advanced AI and machine learning techniques to streamline and enhance social welfare services in Sri Lanka. The Eligibility Prediction Engine utilizes classification models to analyze diverse socio-economic and demographic data—including income levels, age, household size, employment status, and disability conditions—to accurately determine citizens' eligibility for various welfare programs. By automating this assessment, the system removes inefficiencies and subjectivity from manual evaluations, ensuring that aid is directed swiftly and fairly to those who qualify.

Complementing this, the Healthcare Recommendation System employs intelligent ranking algorithms and collaborative filtering approaches to suggest appropriate healthcare facilities and specialists tailored to everyone's health profile and needs. This personalized guidance empowers citizens to access suitable medical care efficiently while aiding government officers in resource management.

The integrated AI platform not only simplifies the welfare application and administration processes but also improves transparency and accessibility by offering real-time eligibility checks, multilingual chatbot support, and dynamic recommendation features. Together, these systems foster social equity and enhance well-being by enabling data-driven, precise distribution of welfare and health services to vulnerable populations.

1.1 Background & Literature Review

Financial inclusion and effective financial guidance have become critical components in addressing economic disparities faced by low-income populations worldwide. According to the World Bank's Global Findex Database[1], about 1.7 billion adults remain unbanked globally, lacking access to basic financial services. This exclusion often results in reduced opportunities for wealth accumulation, limited access to credit, and increased vulnerability to economic shocks. In Sri Lanka, a significant portion of welfare recipients experience similar barriers, as the existing financial systems and advisory services do not sufficiently address their specific needs and constraints[2].

Traditional financial advisory services are typically costly and inaccessible to low-income groups, who require affordable, personalized support to make informed decisions about investments and savings. The advent of digital financial services powered by artificial intelligence (AI) offers a promising solution to bridge this gap[3]. AI-driven financial guidance platforms can democratize access by providing micro-investment recommendations and financial planning tailored to each user's socio-economic background, risk appetite, and income dynamics. These systems leverage data such as income patterns, expense tracking, and eligibility for various government aid programs to deliver individualized and scalable advice[4].

Key benefits of AI-powered financial guidance systems include:

- **Improved Accessibility:** These platforms make financial advice available to users in remote or underserved regions via mobile or web applications, bypassing the need for physical financial advisors[3].
- **Cost Reduction:** Automation lowers service delivery costs, enabling sustainable support for economically disadvantaged groups[4].
- **Personalized Recommendations:** Advanced clustering algorithms segment users into homogeneous groups, enabling more relevant investment options and financial aid schemes tailored to specific profiles[5].
- **Dynamic Adaptation:** Recommender systems constantly update advice based on changing financial situations, allowing users to adjust plans as their circumstances evolve[6].

Despite these advantages, implementing AI in welfare-focused financial guidance raises several challenges:

- **Data Sensitivity and Privacy:** Handling financial and personal data of vulnerable populations requires stringent security and ethical frameworks to protect user confidentiality[7].
- **User Digital Literacy:** Many low-income individuals have limited experience with digital technologies; thus, systems must be intuitive and provide trustworthy human-centered support[8].

- **Algorithmic Fairness:** AI models must be carefully developed and monitored to avoid perpetuating biases or excluding certain user groups due to biased training data or flawed assumptions[9].
- **Integration with Welfare Programs:** Coordination between AI financial guidance and existing welfare schemes is often lacking, reducing effectiveness and uptake among beneficiaries[10].

A review of existing literature reveals growing evidence supporting the role of AI in enhancing financial inclusion:

- The International Finance Corporation (IFC) reported in 2023 that AI-enabled micro-investment platforms increased savings rates by up to 35% in underserved communities across Asia and Africa, demonstrating substantial economic empowerment effects[11].
- Chen et al. (2022) illustrated that clustering algorithms effectively segmented low-income populations for credit risk scoring and personalized investment suggestions, leading to improved financial decision-making accuracy[5].
- Kumar and Singh (2021) found that recommender systems enhanced user engagement and satisfaction when transparent explanations of financial advice were incorporated, building critical trust in AI tools for low-income users[6].

Nonetheless, few platforms explicitly focus on integrating welfare program data and addressing the multi-dimensional needs of welfare recipients[10]. The proposed AI-based financial guidance module uniquely aims to fill this gap within the Sri Lankan social protection ecosystem, supporting users with tailored micro-investment plans and financial aid recommendations that amplify the benefits of digital welfare services.

By synthesizing insights from recent advances in AI, behavioral economics, and social welfare policy, this research sets out to develop a financial guidance system that is not only technically robust but socially inclusive and contextually relevant. The goal is to empower vulnerable citizens with smart, evidence-based financial advice, thus fostering enhanced economic resilience, financial literacy, and improved welfare outcomes.

1.2 Research Gap

A research gap refers to an area within a field of study that has not been adequately explored or addressed by prior studies. In our investigation related to financial guidance for low-income welfare recipients, several key gaps have been identified.

The first significant research gap concerns the lack of personalized, evidence-based financial advice tailored to the unique needs of welfare beneficiaries. Current financial aid programs often allow recipients to self-select financial support options or micro-investment plans based on limited information or personal preference. However, this approach carries risks, as individuals might choose options that seem beneficial but may ultimately worsen their financial stability. For instance, a beneficiary may opt for a high-risk micro-investment appealing in the short term, but if market conditions turn unfavorable, this could increase their financial vulnerability instead of reducing it. This gap underlines the need for intelligent recommendation systems that factor in users' income, expense patterns, and risk profiles to suggest optimized and safer financial plans.

The second gap is the absence of continuous monitoring and feedback mechanisms that track the financial progress and investment outcomes of welfare recipients after advice delivery. Existing systems rarely provide ongoing support or detailed insights into how users' financial behaviors evolve over time. Consequently, advisors and policymakers lack real-time understanding of beneficiaries' financial health post-intervention, limiting their ability to offer timely adjustments or additional guidance. Without such monitoring, the system cannot dynamically update financial recommendations or identify recipients who may be struggling, leading to inefficient use of welfare resources and missed opportunities for economic empowerment.

Addressing these research gaps by integrating AI-powered clustering techniques and recommender systems into a financial guidance module can improve personalization, risk management, and adaptive feedback. This innovation is crucial to enhancing the welfare ecosystem and ensuring low-income citizens receive the most effective and sustainable financial support.

Table 1.0 : Comparison of Key Features Between Proposed AI-Powered Financial Module and Existing Advisory Methods

Feature	Proposed Platform	Current Sri Lanka Welfare Systems (Manual System)
Personalized micro-Investment Recommendations	Yes	No or Limited
Clustering based on Financial Behavior	Yes	No
Continuous based Tracking & Monitoring	Yes	No
Adaptive Follow-up Scheduling	Yes	No
Integration with welfare Program Data	Yes	No
Multilingual Support	Yes	Partial

1.3 Research Problem

The research problem defines the gap in existing knowledge that this study seeks to address. Below are the primary research questions identified concerning the AI-based financial guidance module:

- **How to ensure that the recommended financial plans or micro-investment options effectively improve the financial well-being of low-income welfare recipients?**

The first challenge is to identify and validate financial advice or investment recommendations that genuinely benefit beneficiaries without exposing them to undue risk. The approach involves conducting a thorough review of financial inclusion literature and consulting financial experts specializing in low-income populations. Additionally, engaging with welfare recipients to understand their preferences and financial behaviors is crucial. This ensures that recommendations are both effective and acceptable to users, thereby maximizing real-world applicability. The ultimate goal is to design recommendation algorithms that consistently suggest financial options proven to enhance economic stability for welfare users.

- **How to create reliable methods to monitor and track financial behavior and outcomes of users after recommendations are given?**

The second research problem focuses on developing mechanisms to accurately track beneficiaries' adherence to the recommended plans and monitor their financial progress over time. This involves designing data collection methods and analytic tools to assess measurable impacts, such as saving rates, investment engagement, or improvements in financial literacy. Ensuring the tracking system is effective and user-friendly is critical for maintaining continuous support and enabling timely interventions if users deviate from advised actions.

- **How to accurately predict and adapt follow-up financial guidance or scheduling based on users' progress?**

The third research problem is to create a dynamic, responsive mechanism that adjusts the frequency and type of financial advice or support sessions according to the tracked progress of beneficiaries. Rather than relying on fixed schedules, the system will analyze ongoing user data to predict when a user requires additional engagement or re-evaluation of guidance. For example, if a user is demonstrating consistent progress and adherence, less frequent follow-ups might be scheduled. Conversely, if a user is struggling with adherence or facing financial setbacks, the system recommends earlier or intensified support to prevent worsening financial vulnerability.

2. MAIN AND SUB OBJECTIVES

2.1 Main Objectives

The main objective of this research is to enhance the efficiency and effectiveness of AI-driven financial guidance for low-income welfare recipients in Sri Lanka. This innovation aims to empower beneficiaries by providing personalized financial advice, micro-investment recommendations, and dynamic support based on their financial behaviors and profiles. To deliver a more tailored and adaptive financial empowerment experience, the research emphasizes the design and integration of intelligent clustering algorithms, recommender systems, progress tracking mechanisms, and adaptive follow-up scheduling within a unified digital welfare platform.

To accomplish the above main objective, the following four sub-objectives must be achieved:

- Develop clustering algorithms to segment welfare recipients based on financial backgrounds, income patterns, and risk profiles to facilitate precise targeting of financial recommendations.
- Implement a financial recommendation system that suggests suitable micro-investment plans and aid options customized to individual beneficiary needs and preferences.
- Create a mechanism to monitor and analyze user engagement with the recommended financial plans over time, capturing adherence, usage patterns, and financial outcomes.
- Design an adaptive scheduling module to recommend timely follow-up advisory sessions or interventions based on the tracked progress and changing financial conditions of users.

2.2 Sub Objectives

The core component for which I am responsible involves delivering an AI-driven financial guidance module that recommends appropriate micro-investment and financial aid plans, monitors beneficiary engagement, and schedules subsequent advisory interactions. To achieve this, I have further subdivided my work into the following specific sub-objectives:

- Provide personalized financial plans by using AI to curate tailored, realistic investment and aid options aligned with each low-income user's financial situation and goals.
- Track beneficiary adherence and progress by monitoring their saving habits, investment participation, and aid use to evaluate the guidance's effectiveness
- Deliver actionable insights to officers via diagnostic reports that provide visibility into user behavior and improvement, informing interventions and policy adjustments.
- Predict and schedule follow-ups by using progress data to forecast when a beneficiary needs guidance, prioritizing users in distress over stable ones.

3. METHODOLOGIES

3.1 Overall System

The entire system is designed to enhance the efficiency and effectiveness of social welfare service delivery through AI-powered financial guidance. The system is composed of four major components, each addressing critical aspects of digital financial assistance for low-income welfare recipients:

- **User Financial Profile Clustering:** This component collects and processes anonymized financial data from welfare recipients, including income, expenses, and demographic attributes. Advanced clustering algorithms segment users into meaningful groups based on financial behavior and risk profiles, enabling targeted recommendation strategies.
- **Micro-Investment and Financial Aid Recommendation System:** Leveraging the clusters generated, this component applies collaborative filtering and rule-based techniques to suggest appropriate micro-investment plans and government/NGO financial aid options. The system aims to personalize financial guidance based on individual socio-economic characteristics and preferences.
- **Engagement and Progress Tracking Module:** This component monitors users' adherence and interaction with the recommended financial plans, tracking key metrics such as saving frequency, investment participation, and utilization of aid programs. The collected data supports ongoing evaluation of guidance effectiveness and user behavior.
- **Adaptive Follow-up Scheduling Algorithms:** Using insights from the engagement module, this component predicts optimal timing for follow-up financial advisory sessions or interventions. It dynamically adjusts scheduling based on users' financial progress and emerging needs, ensuring timely support and maximizing the impact of financial guidance.

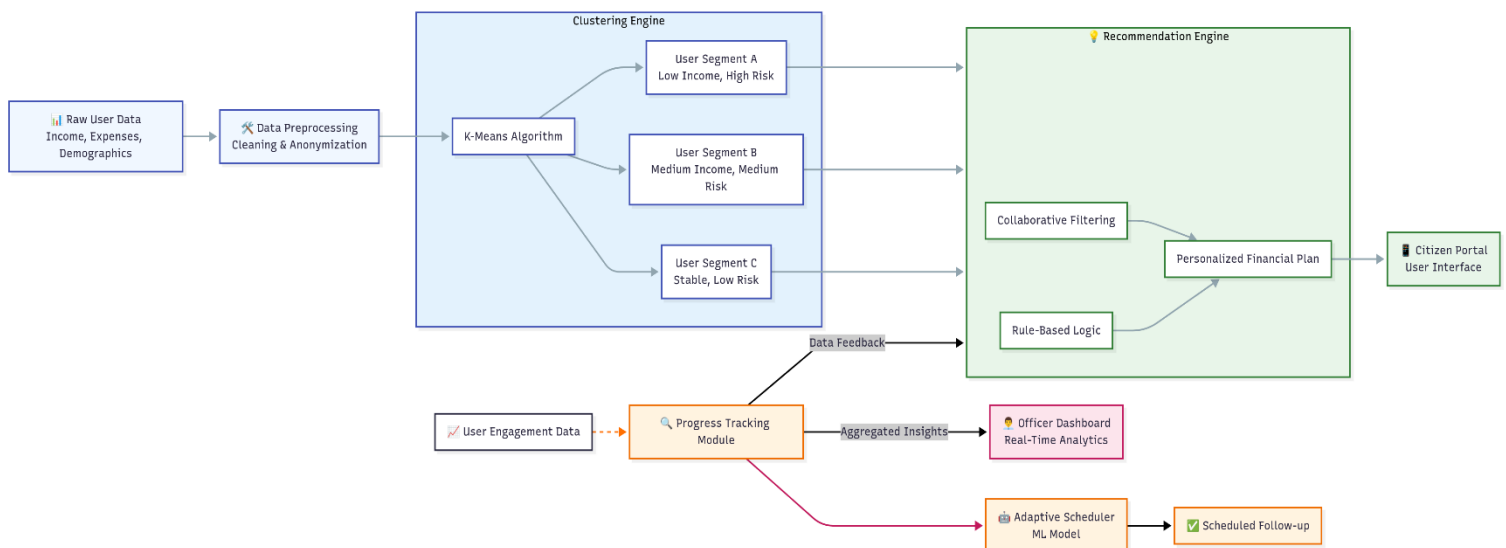
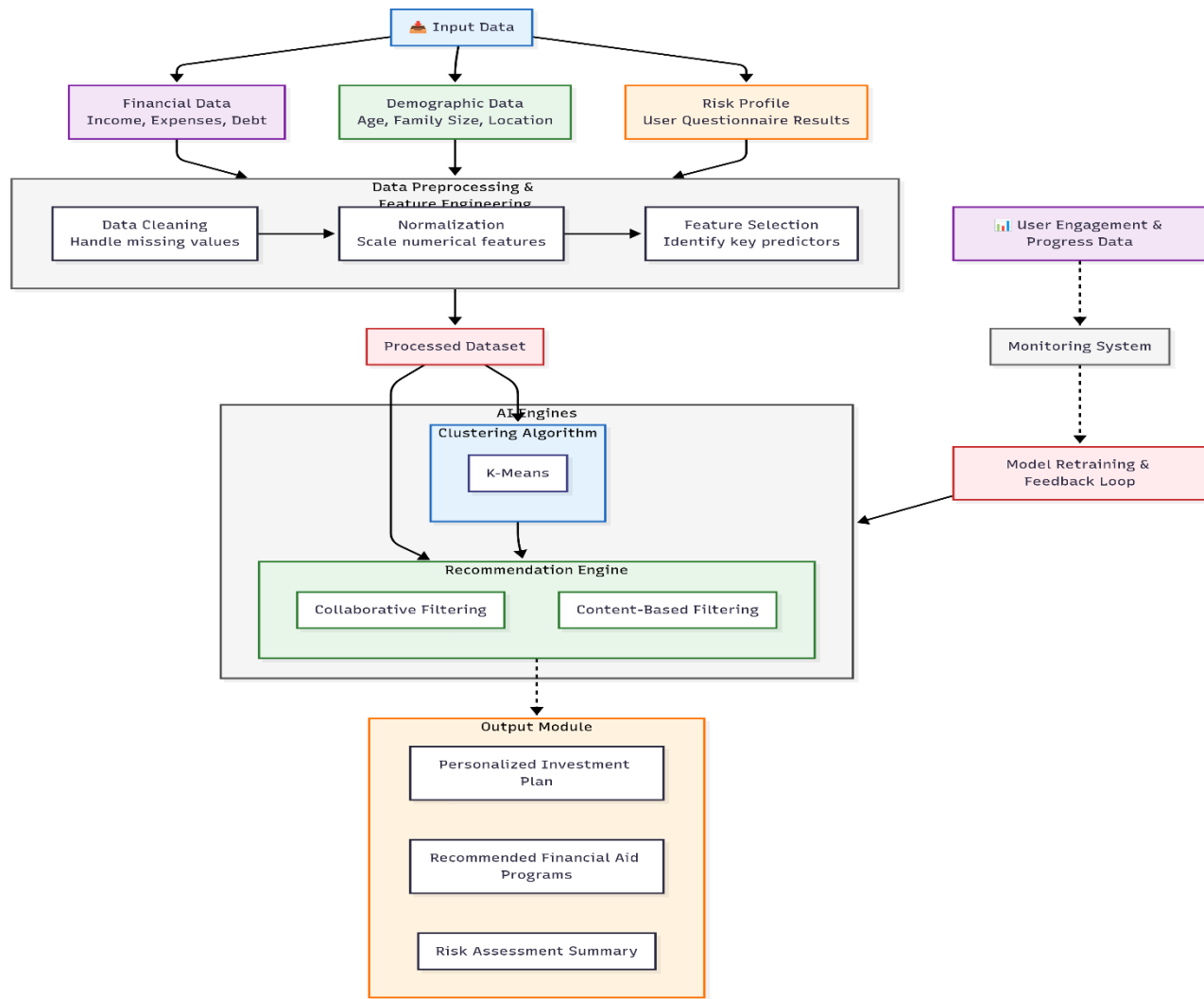


Figure 2: Whole System Diagram

3.2 Individual System Diagram



This component, which I am responsible for developing, integrates financial clustering and personalized recommendation to empower welfare officers and citizens with timely, accurate guidance for economic empowerment.

The system is divided into four core subcomponents:

- **Data Collection and Preprocessing:** This subcomponent gathers and prepares financial and demographic data from multiple sources, including user-inputted income/expense details and existing welfare program records. It ensures data quality, handles missing values, and normalizes features for consistency, creating a reliable dataset for machine learning models.
- **User Clustering Module:** Applies unsupervised machine learning algorithms, primarily K-Means clustering, to segment welfare recipients into distinct groups based on their financial behavior, risk capacity, and economic background. This enables targeted and precise recommendation strategies for each user segment.
- **Financial Recommendation Engine:** Utilizes collaborative filtering and content-based ranking algorithms to analyze a user's cluster and individual profile. It then suggests suitable micro-investment plans and financial aid programs tailored to the beneficiary's specific needs and goals.
- **Integration and Monitoring Interface:** Connects the AI models to the citizen portal and officer dashboards. It provides intuitive access to recommendations for users and delivers real-time feedback and progress tracking for officers, enabling data-driven interventions and support.

4. GANTT CHART AND WORK BREAKDOWN STRUCTURE

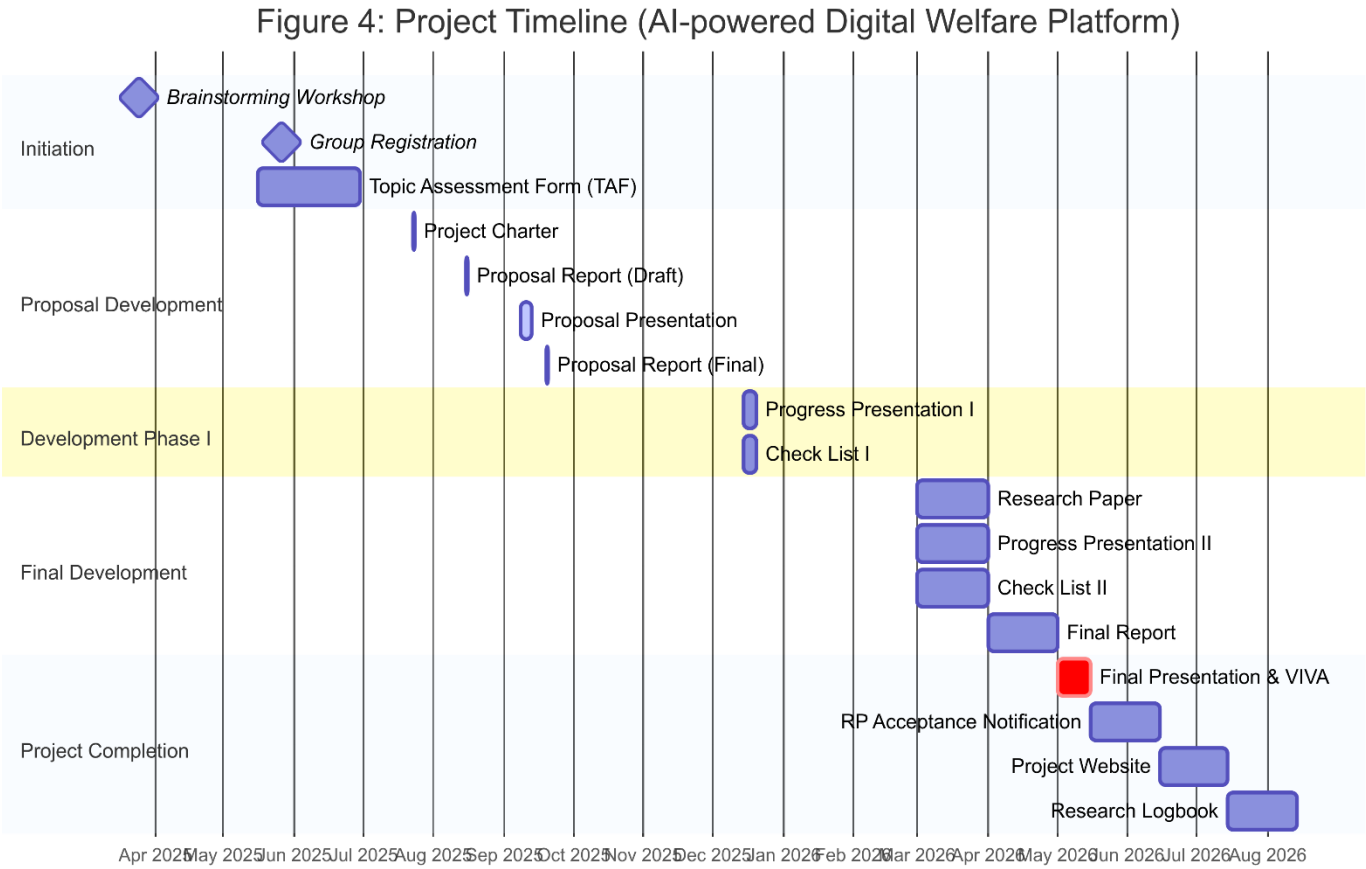


Figure 4: Gantt Chart

5. PROJECT REQUIREMENTS

5.1 Functional Requirements

The precise actions, duties, or responsibilities that the AI-based financial guidance system must carry out are outlined in the functional requirements. The system's functionality, including the inputs it should accept, the outputs it should generate, and the behavior it should display under different conditions, are described in these requirements. Ensuring the system meets user needs is dependent on clearly defined functional requirements.

Functional requirements for the AI-based financial guidance module include:

- **Personalized Financial Plan Recommendation:** The system should allow generation, and assignment of customized micro-investment and financial support plans based on each welfare recipient's financial profile, including income, expenses, and risk tolerance.
- **Progress Tracking:** The system must track and analyze beneficiaries' interactions with the recommended financial plans, including monitoring saving behavior, investment participation, and usage of aid programs. Engagement overtime should be captured in detailed progress reports.
- **Advisor Feedback:** The system should generate diagnostic reports and insights based on user financial activity data, enabling welfare officers and financial advisors to make informed decisions about plan adjustments or additional support.
- **Adaptive Follow-up Scheduling:** Based on the analysis of users' engagement and financial progress, the system should predict and suggest optimal timing for follow-up advisory sessions or interventions to maximize financial empowerment and welfare impact.

5.2 Non-Functional Requirements

Non-functional requirements define system attributes related to quality rather than specific behaviors. For the financial guidance module, these include scalability, performance, security, usability, and reliability.

- **Security:** The system must ensure the confidentiality and integrity of users' sensitive financial data and diagnostic reports, enforcing strict access controls.
- **Reliability:** The system should be highly available and responsive, providing continuous access to guidance and tracking features for beneficiaries and advisors.
- **Efficiency:** The system must efficiently process financial data and user interactions to provide timely, accurate progress tracking and adaptive recommendations.
- **Usability:** The user interface should be intuitive and accessible, designed to cater both to low-income beneficiaries with varying digital literacy levels and welfare officers who monitor and advise them.

6. COMMERCIALIZATION

6.1 Budget

The following budget table provides an estimated cost breakdown for launching and maintaining the proposed AI-Based Investment Plan Recommendation Module. The costs are approximate and may vary depending on vendor pricing, system scale, and chosen service providers..

Description	Cost	Occurring
Cloud Server Hosting (AWS EC2 / Azure VM)	LKR 12,000	Monthly
Mobile App Store Fees (Google Play & Apple Developer)	LKR 38,454	Annual
Financial Data API Subscriptions	LKR 20,000	Monthly
Cloud Database (Firestore / AWS RDs)	25 USD	Monthly
Domain Registration & SSL Certificate	LKR 5,000	Annual
Security & Compliance Audit	LKR 75,000	One-Time
Marketing & User Acquisition	LKR 60,000	One-Time
Maintenance & Technical Support	LKR 30,000	Monthly

Table 2: Budget Breakdown

6.2 Commercialization Methods

The system will be introduced as a scalable fintech solution that can be adopted by government welfare agencies, financial institutions, and non-profit organizations (NGOs). Our commercialization strategy focuses on a B2B (Business-to-Business) model with a B2B2C (Business-to-Business-to-Consumer) approach for end-user delivery.

1. B2B Licensing to Government & Financial Institutions:

- Primary Client: The Sri Lankan government (e.g., Ministry of Finance, Samurdhi Authority). The module would be licensed as a SaaS (Software as a Service) product to be integrated directly into the national digital welfare platform.
- Partnerships with Banks/NGOs: Partner with regulated financial institutions and non-profits offering micro-investment products. Their products are integrated into the recommendation engine, and they pay a referral fee or subscription for access to this targeted user base.

2. Freemium Model for End-Users (Citizens):

- Free Tier: All citizens have access to basic features: financial health assessment, generic savings tips, and information on government aid schemes.
- Premium Tier: For a low monthly subscription (e.g., LKR 500), users unlock personalized micro-investment recommendations, advanced progress analytics, and priority access to financial advisory sessions.

3. Revenue Streams:

- SaaS Licensing Fees: Recurring revenue from the government and institutional partners.
- Subscription Revenue: From citizens opting for premium features.
- Referral Fees: Commission from financial partners for users who invest through recommended plans.
- Anonymized Insights: Selling aggregated, anonymized data on financial trends of low-income groups to research institutions (strictly adhering to data privacy laws like the PDPA).

4. Market Expansion:

- Initially launch as a pilot with a divisional secretariat in Sri Lanka.
- Subsequent rollout aims for nationwide adoption within the government's welfare platform.
- The underlying technology can be white-labeled and adapted for other developing countries with similar socio-economic challenges.

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